Xu He

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EDUCATION

Christ's College, University of Cambridge Sep 2025 - Jul 2026

Master of Mathematics (M.Maths)

- Academic Performance: Part III First with Merit (74/100)

Christ's College, University of Cambridge Sep 2022 - Jul 2025

Bachelor of Arts (BA) in Natural Sciences

- Academic Performance: First Class Throughout; Part IA 71.4/100, Part IB 112/545, Part II 78.8/100 (19/129)

SCHOLARSHIPS AND AWARDS

Greig Prize, Christ's College, University of Cambridge *2025*

College Scholar, Christ's College, University of Cambridge *2022-2026*

EXPERIENCE

Department of Physics, Hong Kong University of Science and Technology Sep 2026 - Aug 2027

Research Assistant in Yilong Han Lab

- Proved that phase networks for systems with local correlations admits universal Gaussian spectrum.
- Collaborated with experimentalists to compute and simulate Langevin dynamics and coupling of hexadic and glassy phases in 2D colloidal disks. Found correlation oscillate in such system.

DAMTP, University of Cambridge Mar - May 2026

Part III Essay on Pattern Formation in Mixture of Active Passive Particles

- Supervised by Prof. Robert Jack. Investigated the breakdown of equilibrium spinodal- binodal relationships in active matter, building upon the reciprocal Cahn-Hilliard model, non- equilibrium extensions, and the active- passive lattice gas. Showed that a generalised free energy enforces a one- to- one correspondence between stationary spinodal and binodal in scalar and reducible multicomponent systems, while identifying parameter regimes where linear instability persists outside global phase separation.

DAMTP, University of Cambridge Jul - Sep 2025

Research Assistant under Prof. Krippendorf

- Applied generative AI and diffusion models to solve inverse problems in string theory flux vacua. Adapted core methodologies from image generation models to reconstruct 8- dimensional flux vectors, implementing and fine-tuning Latent Diffusion Models (LDM) and Continuous Conditional Diffusion Models (CCDM). Developed expertise in model design, hyperparameter optimization, and preprocessing of high- dimensional discrete data, gaining practical experience with PyTorch and advanced neural network architectures.

Theory of Structure Colour of Siler collingwoodi Jul 2023 - Aug 2024

Cooperate with Huize Gao from Zhang Wei Lab, School of Life Sciences, Peking University

- Through simulations and theoretical analysis, identified systematic errors in classical optical models of chitin layers and validated observational data via wavefront and mode analysis.

Institute of Physics, Chinese Academy of Science (CAS) Jul - Aug 2023

International Conference on Physics under Synergetic Extreme Conditions

- Participated in the summer school, gaining exposure to high- pressure (diamond anvil cells) and ultra- low-temperature (10mK) experimental techniques for quantum materials.

SKILLS

Programming: Python (NumPy, SciPy, PyTorch), LaTeX

Machine Learning: Neural Networks (CNNs, VAEs), Diffusion Models (LDM, CCDM)